Yanis Bendehane

Engineering Student in Computer Science | Data Engineering, Optimization and AI

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Summary

Computer science engineering student in an apprenticeship at Thales. Hands-on experience in data warehousing, ETL and industrial data integration, and in optimization algorithms for vehicle routing. Seeking a 17-week research internship abroad in data science, AI or optimization, mid-June to mid-October 2027.

Education

CESI School of Engineering, MEng in Computer Science – Aix-en-Provence, France — Sept 2023 – Sept 2028

- CTI-accredited engineering degree (Master's level): two-year integrated preparatory cycle (2023–2025), then a three-year apprenticeship engineering cycle with Thales (2025–2028)
- Grade A in every course of semester 5; grade A in operations research

Jean Monnet High School, French Baccalaureate in Mathematics and Physics-Chemistry, with honors – Vitrolles, France — Sept 2020 – July 2023

Experience

Apprentice Software Engineer, Thales Services Numériques – Aix-en-Provence, France — Oct 2025 – present

- Develop and test features of a large-scale, mission-critical defense software system in Java on Linux, within a 40-person engineering team spread over three sites
- Work across the software life cycle of a software evolution (requirements analysis, development, testing, documentation) with Subversion configuration management

IT Intern, LyondellBasell – Berre-l'Étang, France — Apr 2025 – June 2025

- Rebuilt a legacy C++ (MFC) program as a standalone C# Windows service that synchronizes tank logistics states between a real-time industrial database (AspenTech IP.21) and SQL Server over ODBC

Academic Projects

Hospital data warehouse (team of 4, Big Data module) — Oct 2026 – present

- Designed a decision-support data warehouse of four star schemas (conformed dimensions, surrogate keys) for hospital activity data
- Built Talend ETL jobs loading PostgreSQL, with pseudonymization of patient identifiers (salted SHA-256)

Vehicle routing with time windows (team of 4, Operations Research module) — Spring 2026

- Modeled an NP-hard VRPTW (capacity, time windows, horizon) and benchmarked four methods on 150 generated instances (10 to 200 customers) against an OR-Tools reference
- Implemented nearest-neighbor + 2-opt, simulated annealing, a genetic algorithm and a Transformer attention model trained with REINFORCE: at 200 customers it stayed 40% from the reference in under 1 s, versus 156% for simulated annealing

Microservices social network (team of 3, Distributed Applications module) — June 2026

- Owned the authentication service (Node.js, Express, PostgreSQL, Sequelize, JWT with refresh tokens, bcrypt) and the infrastructure layer
- Deployed 7 Docker containers with Docker Compose behind an Nginx API gateway validating JWTs before routing

Leadership and Volunteering

Co-organizer, charity football tournament (CESI civic engineering project) — Jan 2024 – June 2026

- Organized three editions with a team of five; proceeds donated to charities chosen by the winning teams (incl. Les Restos du Cœur, ShareTheMeal)

Skills

Programming: Python (NumPy, pandas, NetworkX, Matplotlib), SQL, Java, C#

Data: PostgreSQL, SQL Server, MongoDB, Talend ETL, Hadoop (HDFS, Hive) on Cloudera, star schemas, ODBC, pseudonymization

Methods: Operations research (heuristics, metaheuristics, OR-Tools), reinforcement learning (REINFORCE), attention models, benchmarking

Tools: Linux, Git, Docker, Excel, Power BI

Languages: French (native), English (B2, TOEIC practice test 865/990), Spanish (school level), Arabic (basic)

Certification: ANSSI SecNumacadémie (French national cybersecurity MOOC)